

Geometria Analítica (2024-1) lista 2.

1) a) $\begin{vmatrix} 2 & 1 \\ -1 & 3 \end{vmatrix} \rightarrow \det A = 2 \cdot A_{11} + 1 \cdot A_{12}$

$A_{11} = (-1)^{1+1} \cdot 3 \rightarrow A_{11} = 3 \quad // \quad A_{12} = (-1)^{1+2} \cdot (-2) \rightarrow A_{12} = +2$

$\det A = 2 \cdot 3 + 1 \cdot 2 \rightarrow \boxed{\det A = 10}$

b) $\begin{vmatrix} \sqrt{2} & 3\sqrt{6} \\ 2 & \sqrt{3} \end{vmatrix} \rightarrow \det A = -\sqrt{2} \cdot A_{11} + 2 \cdot A_{21}$

$A_{11} = (-1)^{1+1} \cdot \sqrt{3} \rightarrow A_{11} = +\sqrt{3} \quad // \quad A_{21} = (-1)^{2+1} \cdot 3\sqrt{6} \rightarrow A_{21} = -3\sqrt{6}$

$\det A = -\sqrt{2} \cdot \sqrt{3} + 2 \cdot (-3\sqrt{6})$

$-\sqrt{2} \cdot \sqrt{3} = -\sqrt{6} \quad + \quad -6\sqrt{6} \rightarrow -\sqrt{6} + -6\sqrt{6}$

$1\sqrt{6} + (-6\sqrt{6}) = \boxed{\det A = -5\sqrt{6}}$

c) $\begin{vmatrix} 1 & 0 & 2 \\ 5 & -1 & 1 \\ 1 & 0 & 0 \end{vmatrix} \quad \det A = 1 \cdot A_{31} + 0 \cdot A_{32} + 0 \cdot A_{33}$
 $\det A = 2$

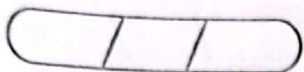
$\begin{vmatrix} 0 & 2 \\ -1 & 1 \end{vmatrix} \quad 0 \cdot 1 - (-2) \rightarrow +2 \rightarrow \boxed{\det A = +2}$

d) $\begin{vmatrix} -2 & 1 & -1 \\ 1 & 5 & 4 \\ -3 & 2 & 2 \end{vmatrix} \rightarrow \det A = -2 \cdot A_{11} + 1 \cdot A_{21} + 3 \cdot A_{31}$
 $(-2 \cdot -6) + (1 \cdot -6) + (3 \cdot 9)$
 $12 + (-6) - 27 \rightarrow \boxed{-21}$

$\begin{vmatrix} -2 & 1 & -1 \\ 1 & 5 & 4 \\ -3 & 2 & 2 \end{vmatrix} \rightarrow \begin{vmatrix} 5 & 4 \\ 2 & 2 \end{vmatrix} \quad 10 - 8 = 2 \quad \left\{ \begin{array}{l} -1^{1+1} \rightarrow +1 \cdot 2 \rightarrow \boxed{2} \\ -1^{2+1} \rightarrow -1 \cdot 6 \rightarrow \boxed{-6} \end{array} \right.$

$\begin{vmatrix} -2 & 1 & -1 \\ 1 & 5 & 4 \\ -3 & 2 & 2 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} \quad 2 - (-2) \rightarrow 4 \quad \left\{ \begin{array}{l} -1^{2+1} \rightarrow -1 \cdot 4 \rightarrow \boxed{-4} \end{array} \right.$

$\det A = \dots$



$$\begin{vmatrix} -2 & 1 & -1 \\ 1 & 5 & 2 \\ -3 & 4 & 2 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & -1 \\ 5 & 2 \end{vmatrix} \rightarrow 2 - (-5) = 7 \quad \left\{ \begin{array}{l} A_{31} = -1^{3+1} \rightarrow +1 \cdot 9 = 9 \end{array} \right.$$

e) $\begin{vmatrix} 0 & 2 & 0 \\ 1 & 3 & 5 \\ 2 & -1 & 2 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 5 \\ 2 & 2 \end{vmatrix} \rightarrow 2 - 10 = -8$

$$A_{12} = (-1)^{1+2} \rightarrow -1 \cdot -8 = 8 \quad \parallel \quad 2 \cdot 8 = 16$$

$$\text{Det } A = 16$$

f) $\begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{vmatrix}$

$$\text{Det} = 3 \cdot A_{11} + 0 \cdot A_{21} + 0 \cdot A_{31} + 0 \cdot A_{41}$$

$$\text{Det} = 3 \cdot 2 + 0 + 0 + 0$$

$$\text{Det} = 6$$

$$\begin{vmatrix} 1 & 0 & 1 & 1 & 0 \\ -1 & 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 & 1 \end{vmatrix} \rightarrow -1 + 0 - 1 - 0 + 0 + 0 = -2$$

$$\det A_{11} = (-1)^{1+1} \rightarrow -1 \cdot (-2) \rightarrow -2$$

g) $\begin{vmatrix} 1 & 0 & 0 & 0 & 0 \\ 5 & 1 & 2 & 5 & 3 \\ 7 & 2 & \sqrt{3} & 0 & 0 \\ 10 & -3 & 6 & 1 & 0 \\ -2 & -3 & 0 & 0 & 0 \end{vmatrix}$

$$\text{Det} = 1 \cdot A_{11} + 0 \cdot A_{12} + 0 \cdot A_{13} + 0 \cdot A_{14} + 0 \cdot A_{15}$$

$$\text{Det} = 1 \cdot 9\sqrt{3} + 0 + 0 + 0 + 0$$

$$\text{Det} = 9\sqrt{3}$$

$$\begin{vmatrix} 1 & 2 & 5 & 1 \\ 2 & \sqrt{3} & 0 & 0 \\ -3 & 6 & 1 & 0 \\ -3 & 0 & 0 & 0 \end{vmatrix}$$

$$\text{Det } B = 3 \cdot A_{11}$$

$$\text{Det } B = 3 \cdot 3\sqrt{3}$$

$$\text{Det } B = 9\sqrt{3}$$

$$\begin{vmatrix} 2 & -\sqrt{3} & 0 & 2 & \sqrt{3} \\ -3 & 6 & -1 & -3 & 6 \\ -3 & 0 & 0 & -3 & 0 \end{vmatrix}$$

$$0 + 3\sqrt{3} + 0 - 0 + 0 + 0 = 3\sqrt{3}$$

$$B_{11} = (-1)^{1+1} \rightarrow -1 \cdot 3\sqrt{3} \rightarrow -3\sqrt{3}$$

h) $\begin{vmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$ $\rightarrow \text{Det } A = 3 \cdot A_{11} + 0 \cdot A_{12} + 0 \cdot A_{13} + 0 \cdot A_{14} + 0 \cdot A_{15}$
 $\text{Det } A = 3 \cdot 8 + 0 + 0 + 0 + 0$
 $\text{Det } A = -2^2 + 0 + 0 + 0$
 $\text{Det } A = -2^2$
 $\text{Det } A = -2^2$

$\begin{vmatrix} 0 & 0 & 0 & -2 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -2 & 0 & 0 & 0 \end{vmatrix}$ $\rightarrow \text{Det } B = -2 \cdot A_{41}$
 $\text{Det } B = -2 \cdot 2^2$
 $\text{Det } B = -8$
 $A_{11} = (-1)^{1+1} \rightarrow 1 \cdot -8 = -8$

$\begin{vmatrix} 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{vmatrix}$ $\rightarrow 0 + 0 - 2^2 - 0 + 0 + 0$
 -2^2
 $B_{11} = (-1)^{1+1} \rightarrow 1 \cdot -2^2 = -2^2$

② a) $\det(A+B) \rightarrow \begin{vmatrix} 3 & -5 & 7 \\ 2 & 2 & 8 \\ 1 & -9 & 6 \end{vmatrix} + \begin{vmatrix} 2 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -2 \end{vmatrix} \rightarrow \begin{vmatrix} 7 & -2 & 14 \\ 3 & 2 & 10 \\ 2 & -8 & 2 \end{vmatrix}$

$\begin{vmatrix} 7 & -2 & 14 & 7 & -2 \\ 3 & 2 & 10 & 3 & 2 \\ 2 & -8 & 2 & 2 & -8 \end{vmatrix} \rightarrow 28 - 80 - 336 - (-12) - 560 + 112$
 $-388 + 2160 = 72$

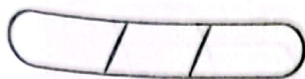
Logo $\det(A+B) = 72$

b) $A = \begin{vmatrix} 3 & -5 & 7 \\ 2 & 2 & 8 \\ 1 & -9 & 6 \end{vmatrix}$ $B = \begin{vmatrix} 2 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -2 \end{vmatrix}$ $\rightarrow \begin{vmatrix} 12+5+21 & 9+0+17 & 21+10-28 \\ 16-2+2^2 & 12+0+8 & 28+2-32 \\ 2+9+18 & 3+0+6 & 7-18-2^2 \end{vmatrix} \rightarrow \begin{vmatrix} 38 & 26 & 3 \\ 38 & 20 & 0 \\ 31 & 9 & -35 \end{vmatrix}$

$\begin{vmatrix} 38 & 16 & -17 \\ 38 & 20 & 0 \\ 31 & 9 & -35 \end{vmatrix}$ $\rightarrow \text{Det} = -17 \cdot (-1)^{1+3} \cdot \begin{vmatrix} 38 & 20 \\ 31 & 9 \end{vmatrix}$
 $342 - 620 = -278$
 $\text{Det} = -17 \cdot 1 \cdot -278$
 $= 4726$

$\text{Det} = -35 \cdot (-1)^{3+3}$ $\rightarrow 4726 + (5320) = -594$

$\begin{vmatrix} 38 & 16 \\ 38 & 20 \end{vmatrix} \rightarrow 1760 - 608 = 1152$
 $\text{Det} = -35 \cdot 1 \cdot 1152 = -39920$
 $\text{Logo } \det(A \cdot B) = -594$



$$c) \text{Det}(B^t \cdot A^t) = \begin{pmatrix} 2 & -1 & 3 \\ 3 & 0 & 1 \\ 7 & 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 & 1 \\ -5 & 2 & -9 \\ 7 & 8 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 12+5+21 & 16-2+24 & 4+9+18 \\ 9+0+7 & 12+0+18 & 3+0+6 \\ 21-10-28 & 28+4-22 & 7+18-24 \end{pmatrix}$$

$$\begin{pmatrix} 38 & 38 & 31 \\ 16 & 20 & 9 \\ -17 & 0 & -35 \end{pmatrix} \begin{pmatrix} 38 & 38 \\ 16 & 20 \\ -17 & 0 \end{pmatrix} \begin{pmatrix} 38 \cdot 20 \cdot 35 + 38 \cdot 9 \cdot -17 + 31 \cdot 16 \cdot 0 \\ -38 \cdot 20 \cdot 35 + 0 \cdot 9 \cdot 38 + 38 \cdot 16 \cdot 35 \\ -26600 - 5817 + 0 \\ -10540 + 0 \\ -21280 \end{pmatrix}$$

$$\text{Logo } \text{Det}(B^t \cdot A^t) = -5924$$

$$d) \text{det}(2A - 3C + B) = 11332$$

$$2A = \begin{pmatrix} 6 & -10 & 14 \\ 8 & 4 & 16 \\ 2 & -18 & 12 \end{pmatrix} + 3C = \begin{pmatrix} -6 & -9 & 3 \\ -18 & -27 & 6 \\ 24 & -36 & 9 \end{pmatrix} + B = \begin{pmatrix} 2 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -1 \end{pmatrix}$$

$$2A - 3C = \begin{pmatrix} 0 & -19 & 17 \\ -10 & -23 & 22 \\ -22 & -57 & 21 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & -19 & 17 \\ -10 & -23 & 22 \\ -22 & -57 & 21 \end{pmatrix} + \begin{pmatrix} 2 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & -16 & 24 \\ -11 & -23 & 24 \\ -19 & -53 & 17 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -16 & 24 \\ -11 & -23 & 24 \\ -19 & -53 & 17 \end{pmatrix} \begin{pmatrix} 2 & -16 \\ -11 & -23 \\ -19 & -53 \end{pmatrix} \begin{pmatrix} 2 \cdot -23 \cdot 17 + (-16) \cdot 24 \cdot (-19) + 24 \cdot (-11) \cdot 53 \\ -(24) \cdot (-23) \cdot (-19) + 2 \cdot 24 \cdot (-53) + (-16) \cdot (-11) \cdot 17 \\ -15674 + 7296 + 13992 = 10488 - 5088 + 2992 \\ 19724 - 8392 = 11332 \end{pmatrix}$$

$$\text{Det}(2A - 3C + B) \rightarrow \text{E igual} = 11332$$

$$e) \text{det}(A \cdot C^t) \rightarrow -11928$$

$$A = \begin{pmatrix} 3 & -5 & 7 \\ 4 & 2 & 8 \\ 1 & -9 & 6 \end{pmatrix} \cdot C^t = \begin{pmatrix} 2 & 6 & 8 \\ 3 & 9 & 12 \\ -1 & -2 & -3 \end{pmatrix} \rightarrow \begin{pmatrix} 6+15-7 & 18-45-17 & 24-72-21 \\ 8+6-8 & 24+18-16 & 32+27-24 \\ 2-27-6 & 6-81-12 & 8-108-18 \end{pmatrix}$$

$$\begin{pmatrix} 14 & -41 & -69 \\ 6 & 26 & 32 \\ -31 & -87 & -118 \end{pmatrix} \begin{pmatrix} 14 & -41 \\ 6 & 26 \\ -31 & -87 \end{pmatrix} \begin{pmatrix} 14 \cdot 26 \cdot -118 + (-41) \cdot 32 \cdot (-31) + (-69) \cdot 6 \cdot (-87) \\ (-69) \cdot 26 \cdot (-31) + (14) \cdot 32 \cdot (-87) + (-41) \cdot 6 \cdot (-118) \\ -212952 + 210672 + 36018 = 33738 \end{pmatrix}$$

$$-55614 - 38976 + 29028 = -75666$$

$$33738 - 75666 = -11928$$



det(A)

③ a) $\det(A^t)$ - Existe uma propriedade de determinantes que diz que:
 $\det A = \det A^t$.

Logo, sabendo q $\det A = -2$, pela propriedade $\det A^t = -2$.

b) $\det(5.A)$

Já que você multiplicou, a determinante não se altera.

Logo $\det(5.A) = -2$

c) $\det(A^6)$

$A = -2$ logo $A^6 = -2 \cdot -2 \cdot -2 \cdot -2 \cdot -2 \cdot -2 = 64$.

Logo $\det(A^6) = 64$

d) $A \cdot A^{-1} = \text{Matriz Identidade}$

Logo $-2 \cdot A^{-1} = 1 \rightarrow A^{-1} = -\frac{1}{2}$

④ a) De acordo com as propriedades: Ao multiplicarmos todos os elementos de uma linha por um número K , o seu determinante fica multiplicado por K , o seu determinante fica multiplicado por K .

Logo $\rightarrow 5 \cdot -3 = -15 = \text{Determinante}$

b)

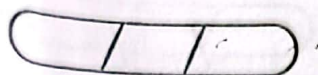
$a \ b \ -2c$	$\rightarrow \div -2 = -2$	$a \ b \ c$	$\rightarrow \text{Determinante dessa matriz} =$
$3d \ 3e \ -6f$		$3d \ 3e \ 3f$	$3 \cdot -3 = -9$
$g \ h \ -2i$		$g \ h \ i$	$-2 \cdot -9 = 18$

Logo a determinante = 18

c)

$-a \ -b \ -c$	$\rightarrow -1 \cdot -1$	$a \ b \ c$	$a \ b \ c$
$g \ h \ i$		$g \ h \ i$	$d \ e \ f$
$-d \ -e \ -f$	$\rightarrow -1 \cdot -1 \cdot -1$	$d \ e \ f$	$g \ h \ i$

$-1 \cdot -3 = 3 \rightarrow \text{Determinante}$



$$d) \begin{vmatrix} G & h & i \\ a & b & c \\ d & e & f \end{vmatrix} \rightarrow -1 \cdot \begin{vmatrix} a & b & c \\ G & h & i \end{vmatrix} \rightarrow -1 \cdot (-1) \cdot \begin{vmatrix} a & b & c \\ d & e & f \end{vmatrix} \rightarrow -1 \cdot (-3) \rightarrow 1 \cdot 3 = -3$$

Determinante = -3

$$e) \begin{vmatrix} a & b & c \\ 2d+a & 2e+b & 2f+c \\ G & h & i \end{vmatrix} = \begin{vmatrix} a & b & c \\ 2d & 2e & 2f \end{vmatrix} + \begin{vmatrix} a & b & c \\ G & h & i \end{vmatrix} \rightarrow \det = 0$$

$$\div 2 \begin{vmatrix} a & b & c \\ d & e & f \\ G & h & i \end{vmatrix} \rightarrow 2 \cdot \det A + 0 = -6$$

f)

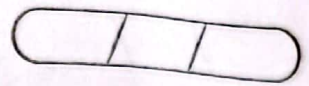
$$\begin{vmatrix} Ka+a & Kb+b & Kc+c \\ d & e & f \\ g & h & i \end{vmatrix} \rightarrow \begin{vmatrix} a & b & c \\ d & e & f \\ G & h & i \end{vmatrix} + \begin{vmatrix} Ka & Kb & Kc \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} \rightarrow \det = 0$$

Logo $-3 + 0 = -3 \rightarrow$ Determinante

5) $\det = -1 + -1 + \div 3 + \div 2$

$$\begin{pmatrix} 5 & 2 & 20 & 1 \\ 4 & 6 & 20 & 4 \\ -5 & -4 & -30 & 9 \\ 3 & -6 & -30 & 12 \end{pmatrix} \xrightarrow{\begin{matrix} \div -1 \\ L_4 + L_3 \end{matrix}} \begin{pmatrix} 5 & 2 & 20 & 1 \\ 4 & 6 & 20 & 4 \\ 5 & 7 & 30 & 9 \\ 8 & 1 & 0 & 3 \end{pmatrix} \xrightarrow{\begin{matrix} L_1 + L_2 \\ \div -1 \end{matrix}} \begin{pmatrix} 1 & -2 & 0 & 5 \\ -4 & -6 & -20 & 4 \\ 5 & 7 & 30 & 9 \\ 8 & 1 & 0 & 3 \end{pmatrix} \begin{matrix} \\ \cdot 3 \\ \cdot 2 \\ \end{matrix}$$





Determinante

$$\begin{pmatrix} 1 & -2 & 0 & 5 \\ -12 & -18 & -60 & 12 \\ 10 & 14 & 60 & -18 \\ 8 & 1 & 0 & 3 \end{pmatrix} \xrightarrow{L_2+L_1} \begin{pmatrix} 1 & -2 & 0 & 5 \\ -2 & -4 & 0 & -6 \\ 10 & 14 & 60 & -18 \\ 8 & 1 & 0 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 0 & 5 \\ -2 & -4 & 0 & -6 \\ 10 & 14 & 60 & -18 \\ 8 & 1 & 0 & 3 \end{pmatrix} \xrightarrow{\begin{matrix} R_2 \div 2 \\ R_3 \div 2 \\ R_4 \div 2 \end{matrix}} \begin{pmatrix} 1 & -2 & 0 & 5 \\ -1 & -2 & 0 & -3 \\ 5 & 7 & 30 & -9 \\ 4 & 1 & 0 & 3 \end{pmatrix}$$

$$-12 + 96 - 10 - (-160) - 6 + 12 = 74 - (-154) \rightarrow 228.$$

$$228 \cdot 60 = 13680 \parallel 13680 \div 2 = 6840 \parallel 6840 \div 3 = 2280$$

$$2280 \cdot (-1) \cdot (-1) \cdot 1 \rightarrow 2280.$$

6 a) $\begin{vmatrix} 4 & 6 & x & 4 & 6 \\ 7 & 4 & 2x & 7 & 4 \\ 5 & 2 & x & 5 & 2 \end{vmatrix} \rightarrow \begin{matrix} -16x + 60x + 14x - 20x + 16x - 72x \\ 58x + 6x = -128 \\ 64x = -128 \rightarrow x = -2. \end{matrix}$

O valor de $x = -2$

6 b) $\begin{vmatrix} 3 & 5 & 7 & 3 & 5 \\ 2x & x & 3x & 2x & x \\ 4 & 6 & 7 & 4 & 6 \end{vmatrix} \rightarrow \begin{matrix} 21x + 60x + 84x - 70x + 54x + 28x \\ 165x - 152x = 39 \\ 13x = 39 \rightarrow 3 = x \end{matrix}$

O valor de $x = 3$.

6 c) $\begin{vmatrix} x+3 & x+1 & x+4 & x+3 & x+1 \\ 4 & 5 & 3 & 4 & 5 \\ 9 & 10 & 4 & 9 & 10 \end{vmatrix} \rightarrow \begin{matrix} 35(x+3) + 27(x+1) + 40(x+4) - 75(x+4) + 30(x+3) + 28(x+1) \\ \rightarrow 35x + 105 + 27x + 27 + 40x + 160 - 75x - 300 + 30x + 90 + 28x + 28 = -7. \\ 102x + 292 - 103x - 298 = -7 \\ -x = -1 \rightarrow x = 1 \end{matrix}$

O valor de $x = 1$

6 d) $\begin{vmatrix} x & x+2 \\ 1 & x \end{vmatrix} = 0 \rightarrow \begin{matrix} x \cdot x - x + 2 \\ x^2 - x + 2 = 0 \end{matrix} \rightarrow \begin{matrix} \Delta = 1 - 4 \cdot 1 \cdot 2 \\ \Delta = -7 \end{matrix} \rightarrow \frac{1 \pm \sqrt{-7}}{2} = \frac{1 \pm i\sqrt{7}}{2}$

$$\frac{1-3}{2} \rightarrow -1$$

$x = 2$ e -1

$$P) \begin{pmatrix} x-1 & 0 & 3 \\ 2 & 0 & x-5 \\ 0 & 3 & 0 \end{pmatrix} \begin{pmatrix} x^{-1} & 0 \\ 2 & 0 \\ 0 & 3 & 1 \end{pmatrix} \rightarrow \begin{matrix} 0+0+18 - (3x-7)(x+108) \\ 18-3x+39x+108 \\ -3x+39x-90 \end{matrix} \rightarrow \begin{matrix} X+13X-30 \neq 0 \end{matrix}$$

$$-13 \pm \sqrt{169-4 \cdot 1 \cdot 30} \quad -13 \pm 7 = 3 \quad -13 \pm 7 = 10$$

Logo X pode ser qualquer número sem ser 3 e 10

7) a) 1º Calcula-se o determinante de A .

$$\text{Det } A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \rightarrow a \cdot d - b \cdot c$$

2º Calcula-se a matriz adjunta de A .

$$A_{11} = (-1)^{1+1} \cdot d \rightarrow d \quad A_{12} = (-1)^{1+2} \cdot b \rightarrow -b$$

$$A_{21} = (-1)^{2+1} \cdot c \rightarrow -c \quad A_{22} = (-1)^{2+2} \cdot a \rightarrow a$$

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

3º Encontre a matriz adjunta

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$4º A^{-1} = \frac{1}{a \cdot d + b \cdot c} \cdot \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$b) \text{Det } A = 3 \cdot 2 - 5 \cdot 1 = 1 \rightarrow A^{-1} = \frac{1}{1} \cdot \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$

$$A_{11} = (-1)^{1+1} \cdot 2 = 1 \cdot 2 = 2$$

$$A_{12} = (-1)^{1+2} \cdot 5 = -1 \cdot 5 = -5$$

$$A_{21} = (-1)^{2+1} \cdot 1 = -1 \cdot 1 = -1$$

$$A_{22} = (-1)^{2+2} \cdot 3 = 1 \cdot 3 = 3$$

$$A^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$

tilibra

$$B^{-1} = B^{-1} = \frac{1}{\det B} \cdot \text{Adj } B \rightarrow \text{Determinante de } B = 1$$

$$A_{11} = (-1)^{1+1} \cdot 2 = 2$$

$$A_{12} = (-1)^{1+2} \cdot 1 = -1 \rightarrow \begin{pmatrix} 2 & -1 \\ -7 & 14 \end{pmatrix} \rightarrow \begin{pmatrix} 2 \cdot 4 \\ -1 \cdot 4 \end{pmatrix} \rightarrow B^{-1} = \frac{1}{1} \cdot \begin{pmatrix} 2 & -7 \\ -1 & 4 \end{pmatrix}$$

$$A_{21} = (-1)^{2+1} \cdot 7 = -7$$

$$A_{22} = (-1)^{2+2} \cdot 4 = 4$$

$$A \cdot B^{-1} = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & -7 \\ -1 & 4 \end{pmatrix} \rightarrow AB = \begin{pmatrix} 12+1 & 21+4 \\ 20+2 & 35+4 \end{pmatrix} = \begin{pmatrix} 13 & 25 \\ 22 & 39 \end{pmatrix} \quad \begin{matrix} 13 \cdot 39 - 22 \cdot 25 = \\ 507 - 550 = -43 \end{matrix}$$

$$A_{11} = (-1)^{1+1} \cdot 39 = 39$$

$$A_{12} = (-1)^{1+2} \cdot 22 = -22 \quad \begin{pmatrix} 39 & -22 \\ -23 & 13 \end{pmatrix} \rightarrow \begin{pmatrix} 39 & -23 \\ -22 & 13 \end{pmatrix}$$

$$A_{21} = (-1)^{2+1} \cdot 23 = -23$$

$$A_{22} = (-1)^{2+2} \cdot 13 = 13$$

$$AB^{-1} = \frac{1}{-43} \cdot \begin{pmatrix} 39 & -23 \\ -22 & 13 \end{pmatrix} \rightarrow AB^{-1} = \begin{pmatrix} 39 & -23 \\ -22 & 13 \end{pmatrix}$$

8) a) $a_{11} = (-1)^{1+1} \cdot 1 = 1$

$$a_{12} = (-1)^{1+2} \cdot 3 = -3 \quad \begin{pmatrix} 1 & -3 \\ 2 & 2 \end{pmatrix}$$

$$a_{21} = (-1)^{2+1} \cdot 2 = -2$$

$$a_{22} = (-1)^{2+2} \cdot 2 = 2$$

b)

$$a_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} \rightarrow (-1)^{1+1} \cdot (2 \cdot (-1) - 1 \cdot 1) = -3$$

$$a_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 1 & 1 \\ 0 & -1 \end{vmatrix} \rightarrow (-1)^{1+2} \cdot (1 \cdot (-1) - 0 \cdot 1) = 1$$

$$a_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \rightarrow (-1)^{1+3} \cdot (1 \cdot 1 - 0 \cdot 2) = 1$$

$$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} -2 & 0 \\ 1 & -1 \end{vmatrix} \rightarrow (-1)^{2+1} \cdot 2 = -1 \cdot 2 = -2$$

$$A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 2 & 0 \\ 0 & -1 \end{vmatrix} \rightarrow (-1)^{2+2} \cdot -2 = 1 \cdot -2 = -2$$

$$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix} \rightarrow (-1)^5 \cdot 2 = -1 \cdot 2 = -2$$

$$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} -2 & 0 \\ 2 & -1 \end{vmatrix} \rightarrow (-1)^4 \cdot -2 = 1 \cdot -2 = -2$$

$$A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} \rightarrow (-1)^5 \cdot 2 = -1 \cdot 2 = -2$$

$$A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 2 & -2 \\ 1 & 2 \end{vmatrix} \rightarrow (-1)^6 \cdot 6 = 1 \cdot 6 = 6$$

$$-3 \quad 1 \quad 1$$

$$-2 \quad -2 \quad -2$$

$$-2 \quad -2 \quad 6$$

9) a) $\begin{pmatrix} 2 & -2 \\ 3 & 1 \end{pmatrix}$ $A_{11} = (-1)^{1+1} = 1 \cdot 1 = 1$ $A_{21} = (-1)^{2+1} = -1 \cdot -2 = 2$
 $A_{12} = (-1)^{1+2} = -1 \cdot 3 = -3$ $A_{22} = (-1)^{2+2} = 1 \cdot 2 = 2$

Det de A = $2 - (-6) = 8$ $\begin{pmatrix} 1 & -3 \\ 2 & 2 \end{pmatrix}$ $\begin{pmatrix} 1 & 2 \\ -3 & 2 \end{pmatrix}$ = matriz adjunta, no 10

$A^{-1} = \frac{1}{8} \cdot \begin{pmatrix} 1 & 2 \\ -3 & 2 \end{pmatrix} \rightarrow A^{-1} = \begin{pmatrix} 1/8 & 1/4 \\ -3/8 & 1/4 \end{pmatrix}$ = matriz inversa

b)

Det de B = $\begin{vmatrix} 2 & -2 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & -1 \end{vmatrix} \begin{vmatrix} 2 & -2 \\ 1 & 2 \end{vmatrix} \rightarrow -4 + 0 + 0 - 0 + 2 + 2 = -8$

$B^{-1} = \frac{1}{-8} \cdot \begin{pmatrix} -3 & -2 & -2 \\ 1 & -2 & -2 \\ 1 & -2 & 6 \end{pmatrix} \rightarrow B^{-1} = \begin{pmatrix} 3/8 & 1/4 & 1/4 \\ -1/8 & 1/4 & 1/4 \\ -1/8 & 1/4 & -3/4 \end{pmatrix}$



$$c) \begin{pmatrix} 0 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 2 & 0 \\ 1 & 1 \end{pmatrix} \rightarrow \text{Det} = 3$$

$$A_{11} = 0 \quad \left\{ \begin{array}{l} A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} \rightarrow -1 \cdot 1 = -1 \\ A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} \rightarrow 1 \cdot 2 = 2 \end{array} \right.$$

$$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} -1 & 1 \\ 0 & -1 \end{vmatrix} = -1 \cdot (-1) = 1 \quad A_{22} = 0 \quad A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} = -1 \cdot 1 = -1$$

$$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} -1 & 1 \\ 0 & -1 \end{vmatrix} = 1 \cdot 1 = 1 \quad A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 0 & 1 \\ 2 & -1 \end{vmatrix} = -1 \cdot 2 = -2 \quad A_{33} = 0$$

$$\begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 1 & 2 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{pmatrix}$$

$$d) \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 2 & 3 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad \text{Det} = \cancel{A_{12}} \cdot 0 + \cancel{A_{22}} \cdot 1 + \cancel{A_{32}} \cdot 0 + \cancel{A_{42}} \cdot 0$$

$$A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & -1 & 0 \end{vmatrix} \rightarrow 1 \cdot (0 \cdot 0 - (-3)) = 3$$

$$A_{12} = 0$$

$$A_{13} = 0$$

$$A_{14} = (-1)^{1+4} \cdot \begin{vmatrix} 0 & 1 & 0 \\ 2 & 0 & 2 \\ 0 & 0 & -1 \end{vmatrix} \rightarrow -1 \cdot (0 \cdot 0 - (-2)) = -2$$

$$A_{21} = 0$$

$$A_{22} = (-1)^{20} \cdot \begin{pmatrix} 1 & 0 & 1 & | & 1 & 0 \\ 2 & 2 & 3 & | & 2 & 2 \\ 0 & -1 & 0 & | & 0 & -1 \end{pmatrix} \rightarrow \begin{matrix} 0+0-2-(0-3+0) \\ -2+3=1. \end{matrix}$$

$$A_{23} = 0$$

$$A_{24} = 0$$

$$A_{31} = (-1)^{3+1} \cdot \begin{pmatrix} 0 & 0 & 1 & | & 0 & 0 \\ 1 & 0 & 0 & | & 1 & 0 \\ 0 & -1 & 0 & | & 0 & -1 \end{pmatrix} \rightarrow \begin{matrix} 0 \cdot 0 - 1 - (0 + 0 + 0) \\ 1 \cdot -1 = -1 \end{matrix}$$

$$A_{32} = 0$$

$$A_{33} = (-1)^6 \cdot \begin{pmatrix} 1 & 0 & 1 & | & 1 & 0 \\ 0 & 1 & 0 & | & 0 & 1 \\ 0 & 0 & 0 & | & 0 & 0 \end{pmatrix} \rightarrow \begin{matrix} 0+0+0-(0+0+0) \\ -1 \cdot 0 = 0 \end{matrix}$$

$$A_{34} = (-1)^7 \cdot \begin{pmatrix} 1 & 0 & 0 & | & 1 & 0 \\ 0 & 1 & 0 & | & 0 & 1 \\ 0 & 0 & -1 & | & 0 & 0 \end{pmatrix} \rightarrow \begin{matrix} -1+0+0-(0+0+0) \\ -1 \cdot -1 = 1. \end{matrix}$$

$$A_{41} = 0 \cdot -2$$

$$A_{42} = 0$$

$$A_{43} = (-1)^7 \cdot \begin{pmatrix} 1 & 0 & 1 & | & 1 & 0 \\ 0 & 1 & 0 & | & 0 & 1 \\ 2 & 0 & 3 & | & 2 & 0 \end{pmatrix} \rightarrow \begin{matrix} 3+0+0-(2+0+0) \\ 3-2=1 \\ -1 \cdot 1 = -1. \end{matrix}$$

$$A_{44} = 2$$

$$\begin{array}{ccc}
 \begin{pmatrix} 3 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 \\ -2 & 0 & -1 & 2 \end{pmatrix} & \rightarrow & \begin{pmatrix} 3 & 0 & -1 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ -2 & 0 & 1 & 2 \end{pmatrix} \quad D = \frac{1}{1} \cdot \begin{pmatrix} 3 & 0 & -1 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ -2 & 0 & 1 & 2 \end{pmatrix}
 \end{array}$$

10 a) $A^t - BX = 2C \rightarrow .1$

2. $-A^t + BX = -2C$

3. $BX = -2C + A^t \rightarrow (B^{-1} \cdot B) \cdot X = B^{-1} \cdot (-2C + A^t)$

4. $1n \cdot X = B^{-1} \cdot (-2C + A^t) \rightarrow X = B^{-1} \cdot (-2C + A^t)$

b) $A^t - BX = 2C \rightarrow \begin{pmatrix} -1 & 0 & -2 \\ -2 & 1 & -3 \\ -4 & -1 & -8 \end{pmatrix} \cdot X = \begin{pmatrix} 0 & 2 & 6 \\ 2 & 0 & 4 \\ 6 & 4 & 0 \end{pmatrix} + \begin{pmatrix} 1 & -2 & -6 \\ -2 & 0 & -4 \\ -6 & 4 & 1 \end{pmatrix}$

$\begin{pmatrix} -1 & 0 & -2 \\ -2 & 1 & -3 \\ -4 & -1 & -8 \end{pmatrix} \cdot X = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \text{Logo.}$

$\begin{pmatrix} -1 & 0 & -2 \\ -2 & 1 & -3 \\ -4 & -1 & -8 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 8+0-4 & -(8-3+0) \\ 4-5 & = -1 \end{pmatrix}$

$A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 1 & -3 \\ -1 & -8 \end{vmatrix} = 1 \cdot 11 = 11$ $A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} -2 & -3 \\ -4 & -8 \end{vmatrix} = -1 \cdot 4 = -4$ $A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} -2 & 1 \\ -4 & -1 \end{vmatrix} = 1 \cdot 6 = 6$

$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} 0 & -2 \\ -1 & -8 \end{vmatrix} = -1 \cdot -2 = 2$ $A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} -1 & -2 \\ -4 & -8 \end{vmatrix} = 1 \cdot 0 = 0$ $A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} -1 & 0 \\ -4 & -1 \end{vmatrix} = -1 \cdot 1 = -1$

$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} 0 & -2 \\ 1 & -3 \end{vmatrix} = 1 \cdot 2 = 2$ $A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} -1 & -2 \\ -2 & -3 \end{vmatrix} = -1 \cdot 1 = 1$ $A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} -1 & 0 \\ -2 & 1 \end{vmatrix} = 1 \cdot 1 = 1$

$B^{-1} = \frac{1}{-1} \cdot \begin{pmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{pmatrix} \rightarrow B^{-1} = \begin{pmatrix} 11 & -2 & -2 \\ 4 & 0 & -1 \\ -6 & 1 & 1 \end{pmatrix} \rightarrow B^{-1} = X \rightarrow \text{Logo } X =$